

CM0845 Logic

Propositional Logic: Natural Deduction

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Preliminaries

Convention

The references for this section are van Dalen [2013, § 2.4 and § 2.6]

Derivation Rules for $\{\wedge, \rightarrow, \perp\}$

Conjunction

$$\frac{\varphi \quad \psi}{\varphi \wedge \psi} \wedge I$$

$$\frac{\varphi \wedge \psi}{\varphi} \wedge E$$

$$\frac{\varphi \wedge \psi}{\psi} \wedge E$$

Derivation Rules for $\{\wedge, \rightarrow, \perp\}$

Implication

$$\frac{\begin{array}{c} [\varphi]^x \\ \vdots \\ \psi \end{array}}{\varphi \rightarrow \psi} \rightarrow I^x \qquad \frac{\varphi \quad \varphi \rightarrow \psi}{\psi} \rightarrow E$$

Remark: In the application of the $\rightarrow I$ rule, we may discharge zero, one, or more occurrences of the assumption.

Derivation Rules for $\{\wedge, \rightarrow, \perp\}$

Bottom elimination

$$\frac{\perp}{\varphi} \perp\text{E}$$

Proof by contradiction (*reductio ad absurdum*)

$$[\neg\varphi]^x$$

$$\vdots$$

$$\frac{\perp}{\varphi} \text{RAA}^x$$

where $\neg\varphi := \varphi \rightarrow \perp$.

Derivation Rules for $\{\wedge, \rightarrow, \perp\}$

Definition

Let Γ be a set of formulae and let φ be a formula. The relation $\Gamma \vdash \varphi$ means that there is a derivation with conclusion φ from the set of hypotheses Γ .

Derivation Rules for $\{\wedge, \rightarrow, \perp\}$

Example

A derivation where every assumption is discharged once. A proof of Pierce's law
 $\vdash ((\varphi \rightarrow \psi) \rightarrow \varphi) \rightarrow \varphi$.*

Proof

$$\frac{\frac{\frac{[\varphi]^x}{\varphi \rightarrow \psi} \rightarrow I^x \quad \frac{\frac{\perp}{\psi} \perp E}{\perp} \rightarrow E}{\frac{[\varphi]^x \quad [\neg\varphi]^y}{\perp} \rightarrow E} \rightarrow E \quad \frac{\frac{((\varphi \rightarrow \psi) \rightarrow \varphi)^z}{\varphi} \rightarrow E \quad [\neg\varphi]^y}{\frac{\perp}{\varphi} \text{RAA}^y} \rightarrow E}{(\varphi \rightarrow \psi) \rightarrow \varphi} \rightarrow I^z$$

*Adapted from [Alastair 2018].

Derivation Rules for $\{\wedge, \rightarrow, \perp\}$

Example

A derivation using the same assumption twice. A proof that $\vdash (\varphi \wedge \psi) \rightarrow (\psi \wedge \varphi)$.

Proof

$$\frac{\frac{\frac{[\varphi \wedge \psi]^x}{\psi} \wedge E}{\psi \wedge \varphi} \wedge I}{(\varphi \wedge \psi) \rightarrow (\psi \wedge \varphi)} \rightarrow I^x$$

Derivation Rules for $\{\wedge, \rightarrow, \perp\}$

Example

A derivation where the assumption and the conclusion are the same. A proof that $\vdash \varphi \rightarrow \varphi$.

Proof

$$\frac{[\varphi]^x}{\varphi \rightarrow \varphi} \rightarrow I^x$$

Derivation Rules for $\{\wedge, \rightarrow, \perp\}$

Remark

'The rule schemes of natural deduction display only the open assumptions that are **active** in the rule, but there may be any number of other assumptions.' [Negri and von Plato 2008, p. 10]

Derivation Rules for $\{\wedge, \rightarrow, \perp\}$

Remark

'The rule schemes of natural deduction display only the open assumptions that are **active** in the rule, but there may be any number of other assumptions.' [Negri and von Plato 2008, p. 10]

Example

A derivation where there is a vacuous discharge when using the inference rule $\rightarrow I$. A proof that $\vdash \varphi \rightarrow (\psi \rightarrow \varphi)$.

Proof

$$\frac{\frac{[\varphi]^x}{\psi \rightarrow \varphi} \rightarrow I \text{ (vacuous discharge of } \psi)}{\varphi \rightarrow (\psi \rightarrow \varphi)} \rightarrow I^x$$

Derivation Rules for $\{\wedge, \rightarrow, \perp\}$

Example

A derivation using one hypothesis. A proof that $\varphi \vdash \neg(\neg\varphi \wedge \psi)$ [van Dalen 2013, Exercise 3.(a), p. 37].

Proof

$$\frac{\varphi \quad \frac{[\neg\varphi \wedge \psi]^x}{\neg\varphi} \wedge\text{E}}{\quad \rightarrow\text{E}} \rightarrow\text{I}^x$$
$$\frac{\perp}{\neg(\neg\varphi \wedge \psi)} \rightarrow\text{I}^x$$

Derivation Rules for $\{\wedge, \rightarrow, \perp\}$

Example

A derivation using the same hypothesis twice. A proof that $\varphi \wedge \psi \vdash \psi \wedge \varphi$.

Proof

$$\frac{\frac{\varphi \wedge \psi}{\psi} \wedge E \quad \frac{\varphi \wedge \psi}{\varphi} \wedge E}{\psi \wedge \varphi} \wedge I$$

Set of Derivations

Notation

(Whiteboard)

Definition (van Dalen [2013], Definition 2.4.1)

The **set of derivations**, denoted $\mathcal{D}$, is the **smallest** set X with the properties:

(see next slide)

Set of Derivations

(1) The one-element tree φ belongs to X for all $\varphi \in PROP$.

(2 $\wedge$) If $\frac{\mathcal{D}}{\varphi}, \frac{\mathcal{D}'}{\varphi'} \in X$, then $\frac{\frac{\mathcal{D}}{\varphi} \quad \frac{\mathcal{D}'}{\varphi'}}{\varphi \wedge \varphi'} \in X$.

If $\frac{\mathcal{D}}{\varphi \wedge \psi} \in X$, then $\frac{\mathcal{D}}{\varphi}, \frac{\mathcal{D}}{\psi} \in X$.

(2 $\rightarrow$) If $\frac{\varphi}{\mathcal{D}} \in X$, then $\frac{[\varphi] \quad \frac{\mathcal{D}}{\psi}}{\varphi \rightarrow \psi} \in X$.

If $\frac{\mathcal{D}}{\varphi}, \frac{\mathcal{D}'}{\varphi \rightarrow \psi} \in X$, then $\frac{\frac{\mathcal{D}}{\varphi} \quad \frac{\mathcal{D}'}{\varphi \rightarrow \psi}}{\psi} \in X$.

(2 $\perp$) If $\frac{\mathcal{D}}{\perp} \in X$, then $\frac{\mathcal{D}}{\varphi} \in X$.

If $\frac{\neg\varphi}{\mathcal{D}} \in X$, then $\frac{[\neg\varphi] \quad \frac{\mathcal{D}}{\perp}}{\varphi} \in X$.

Derivation Rules for the Missing Connectives $\{\vee, \neg, \leftrightarrow\}$

Disjunction

$$\frac{\varphi}{\varphi \vee \psi} \vee\text{I}$$

$$\frac{\psi}{\varphi \vee \psi} \vee\text{I}$$

$$\frac{\begin{array}{c} [\varphi]^x \\ \vdots \\ \varphi \vee \psi \\ \sigma \end{array} \quad \begin{array}{c} [\psi]^y \\ \vdots \\ \sigma \end{array}}{\sigma} \vee\text{E}^{x,y}$$

Derivation Rules for the Missing Connectives $\{\vee, \neg, \leftrightarrow\}$

Negation

$$[\varphi]^x$$
$$\vdots$$
$$\frac{\perp}{\neg\varphi} \neg I^x$$
$$\frac{\varphi \quad \neg\varphi}{\perp} \neg E$$

Derivation Rules for the Missing Connectives $\{\vee, \neg, \leftrightarrow\}$

Equivalence

$$\begin{array}{c} [\varphi]^x \quad [\psi]^y \\ \vdots \quad \vdots \\ \frac{\psi \quad \varphi}{\varphi \leftrightarrow \psi} \leftrightarrow \text{I}^{x,y} \end{array}$$

$$\frac{\varphi \quad \varphi \leftrightarrow \psi}{\psi} \leftrightarrow \text{E}$$

$$\frac{\psi \quad \varphi \leftrightarrow \psi}{\varphi} \leftrightarrow \text{E}$$

Derivation Rules for $\{\wedge, \vee, \rightarrow, \perp\}$

$$\frac{\varphi \quad \psi}{\varphi \wedge \psi} \wedge I$$

$$\frac{\varphi \wedge \psi}{\varphi} \wedge E$$

$$\frac{\varphi \wedge \psi}{\psi} \wedge E$$

$$\frac{\varphi}{\varphi \vee \psi} \vee I$$

$$\frac{\psi}{\varphi \vee \psi} \vee I$$

$$\frac{\begin{array}{c} [\varphi]^x \\ \vdots \\ \varphi \vee \psi \\ \sigma \end{array} \quad \begin{array}{c} [\psi]^y \\ \vdots \\ \sigma \end{array}}{\sigma} \vee E^{x,y}$$

$$\frac{\begin{array}{c} [\varphi]^x \\ \vdots \\ \psi \end{array}}{\varphi \rightarrow \psi} \rightarrow I^x$$

$$\frac{\varphi \quad \varphi \rightarrow \psi}{\psi} \rightarrow E$$

$$\frac{\perp}{\varphi} \perp E$$

$$\frac{\begin{array}{c} [\neg\varphi]^x \\ \vdots \\ \perp \end{array}}{\varphi} \text{RAA}^x$$

Derivation Rules for $\{\wedge, \vee, \rightarrow, \perp\}$

Example

Prove that $\vdash \varphi \vee \neg\varphi$ [van Dalen 2013, example p. 49].

Proof

$$\frac{\frac{\frac{[\varphi]^x}{\varphi \vee \neg\varphi} \vee\text{I} \quad [\neg(\varphi \vee \neg\varphi)]^y}{\rightarrow\text{E}} \quad \frac{\frac{\perp}{\neg\varphi} \rightarrow\text{I}^x}{\varphi \vee \neg\varphi} \vee\text{I} \quad [\neg(\varphi \vee \neg\varphi)]^y}{\rightarrow\text{E}} \quad \frac{\perp}{\varphi \vee \neg\varphi} \text{RAA}^y$$

Natural Deduction in Sequent Calculus Style

$$\frac{}{\Gamma, \varphi \vdash \varphi} \text{Ax}$$

$$\frac{\Gamma \vdash \varphi \quad \Gamma \vdash \psi}{\Gamma \vdash \varphi \wedge \psi} \wedge \text{I}$$

$$\frac{\Gamma \vdash \varphi \wedge \psi}{\Gamma \vdash \varphi} \wedge \text{E}$$

$$\frac{\Gamma \vdash \varphi \wedge \psi}{\Gamma \vdash \psi} \wedge \text{E}$$

$$\frac{\Gamma \vdash \varphi}{\Gamma \vdash \varphi \vee \psi} \vee \text{I}$$

$$\frac{\Gamma \vdash \psi}{\Gamma \vdash \varphi \vee \psi} \vee \text{I}$$

$$\frac{\Gamma \vdash \varphi \vee \psi \quad \Gamma, \varphi \vdash \sigma \quad \Gamma, \psi \vdash \sigma}{\Gamma \vdash \sigma} \vee \text{E}$$

$$\frac{\Gamma, \varphi \vdash \psi}{\Gamma \vdash \varphi \rightarrow \psi} \rightarrow \text{I}$$

$$\frac{\Gamma \vdash \varphi \quad \Gamma \vdash \varphi \rightarrow \psi}{\Gamma \vdash \psi} \rightarrow \text{E}$$

$$\frac{\Gamma \vdash \perp}{\Gamma \vdash \varphi} \perp \text{E}$$

$$\frac{\Gamma, \neg \varphi \vdash \perp}{\Gamma \vdash \varphi} \text{RAA}$$

Natural Deduction in Sequent Calculus Style

Example

We prove that $\vdash \varphi \vee \neg\varphi$.

(continued on next slide)

Natural Deduction in Sequent Calculus Style

Proof

Let $\Gamma = \{\varphi, \neg(\varphi \vee \neg\varphi)\}$ and $\Delta = \Gamma - \{\varphi\}$.

$$\frac{\frac{\frac{}{\Gamma \vdash \varphi} \text{Ax}}{\Gamma \vdash \varphi \vee \neg\varphi} \vee\text{I} \quad \frac{}{\Gamma \vdash \neg(\varphi \vee \neg\varphi)} \text{Ax}}{\Gamma \vdash \perp} \rightarrow\text{E}$$
$$\frac{\frac{\Gamma \vdash \perp}{\Delta \vdash \neg\varphi} \rightarrow\text{I}}{\Delta \vdash \varphi \vee \neg\varphi} \vee\text{I}$$
$$\frac{\Delta \vdash \varphi \vee \neg\varphi \quad \frac{}{\Delta \vdash \neg(\varphi \vee \neg\varphi)} \text{Ax}}{\Delta \vdash \perp} \rightarrow\text{E}$$
$$\frac{\Delta \vdash \perp}{\vdash \varphi \vee \neg\varphi} \text{RAA}$$

Natural Deduction in Sequent Calculus Style

Example

A derivation where there is a vacuous discharge when using the inference rule $\rightarrow$ I. A proof that $\vdash \varphi \rightarrow (\psi \rightarrow \varphi)$.

Proof

$$\frac{\frac{\frac{}{\varphi \vdash \varphi} \text{Ax}}{\varphi \vdash \psi \rightarrow \varphi} \rightarrow\text{I (vacuous discharge of } \psi)}{\vdash \varphi \rightarrow (\psi \rightarrow \varphi)} \rightarrow\text{I}$$

References

-  Alastair, Carr (Mar. 2018). Natural Deduction Pack. URL: <https://github.com/Alastair-Carr/Natural-Deduction-Pack> (visited on 06/08/2025) (cit. on p. 7).
-  Negri, Sara and von Plato, Jan [2001] (2008). Structural Proof Theory. Digitally printed version. Cambridge University Press (cit. on pp. 10, 11).
-  van Dalen, Dirk [1980] (2013). Logic and Structure. 5th ed. Springer (cit. on pp. 2, 12, 14, 20).